

Reward and observation: what each component is, and how it is computed

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Reference for the environment's state score φ and its Dict observation. Source of truth: `Environment.compute_state_score`, `Environment.compute_rigidity_features`, `build_dict_obs` and the primitives in `rigidity.py` / `network.py`.

1 Notation and episode constants

Agent i has position $p_i \in \mathbb{R}^3$ and attitude $R_i \in SO(3)$; a *domain* $\mathcal{D}_i \in \{\mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^2 \times S^1, \mathbb{R}^3 \times S^1, SE(3)\}$ fixes which coordinates it can vary. A directed edge $i \rightarrow j$ means i measures the bearing to j :

$$p_{ij} = p_j - p_i, \quad \hat{p}_{ij} = p_{ij} / \|p_{ij}\|, \quad b_{ij} = R_i^\top \hat{p}_{ij}, \quad P(x) = I - xx^\top.$$

The extended bearing rigidity matrix is $\mathbf{B} = [D_p \bar{E}^\top \bar{S} \mid D_a \bar{E}_o^\top \bar{P}] \in \mathbb{R}^{3m \times 6n}$, with one 3-row block per directed edge, S_i and P_i the per-node translational and rotational DOF projectors, and per edge $k = (i, j)$

$$D_{p,k} = \frac{1}{\|p_{ij}\|} R_i^\top P(\hat{p}_{ij}), \quad D_{a,k} = -R_i^\top [\hat{p}_{ij}]_\times.$$

Length normalisation. $\mathbf{B}$'s position columns carry 1/length and its attitude columns are dimensionless, so any spectrum read off $\mathbf{B}$ moves when the formation is rescaled. Everything spectral below is therefore read off

$$\tilde{\mathbf{B}} = \mathbf{B} \cdot \text{diag}(L I_{3n}, I_{3n}), \quad L = \left(\frac{1}{n} \sum_i \|p_i - \bar{p}\|^2 \right)^{1/2}$$

(`scaled_rigidity_matrix`, `characteristic_length`). Column scaling by a positive scalar leaves the rank unchanged, so the rigidity verdict is unaffected.

Episode constants (`compute_episode_constants`; poses only, edge independent, computed once per `reset`).

symbol	meaning	status
$\text{rank}_{\mathcal{K}}$	rank $\mathbf{B}$ of the complete graph on these poses	exact
$c_{\max}$	most rank one edge block can carry, $\max_k \text{rank } \mathbf{B}_{\mathcal{K}}[3k:3k+3, :]$	exact
m_{req}	fewest edges that could make these poses rigid	<i>lower bound</i>
L	RMS radius about the centroid	exact
$Z_{\mathcal{K}}$	orthonormal basis of $\ker \tilde{\mathbf{B}}_{\mathcal{K}}$, the trivial variations	exact

m_{req} is a subadditivity bound, not a ground truth, and is deliberately kept out of the reward. Closed forms, validated exactly on 216 configurations for $n = 4 \dots 24$:

$$\text{rank}_{\mathcal{K}} = \sum_i \dim \mathcal{D}_i - q_t, \quad q_t = \underbrace{(2 \text{ if any planar else } 3)}_{\text{translations}} + \underbrace{1}_{\text{scale}} + q_r,$$

$$q_r = \begin{cases} 0 & \text{some agent is } \mathbb{R}^2 \text{ or } \mathbb{R}^3 \text{ (no frame)} \\ 3 & \text{every agent is SE(3)} \\ 1 & \text{otherwise (all framed, shared } S^1 \text{ axis)} \end{cases} \quad c_k = \begin{cases} 1 & \text{both ends planar} \\ 2 & \text{otherwise,} \end{cases}$$

with $\dim \mathcal{D}_i \in \{2, 3, 3, 4, 6\}$ for the five domains in the order above, planar meaning $\mathbb{R}^2$ or $\mathbb{R}^2 \times S^1$, and m_{req} the greedy accumulation of the c_k sorted descending until rank_K is reached.

2 Reward

The step reward is potential based,

$$r_t = \varphi(s_{t+1}) - \varphi(s_t) - \text{time_penalty_value} + [\text{terminal bonus}],$$

so it is the *improvement* in φ , not its level. The discount must stay $\gamma < 1$: at $\gamma = 1$ the return telescopes to $\varphi(s_T) - \varphi(s_0)$ and the advantage collapses to ≈ 0 under a mixing policy.

WeightedNormalizedShapeError (**current**)

$$\varphi = \frac{w_r \text{rank } \mathbf{B} - w_e m c_{\max}}{\text{rank}_K} + \underbrace{\kappa \cdot \frac{w_e c_{\max}}{\text{rank}_K}}_{\text{one edge}} \cdot \mathbb{I}[\text{IBR}] \cdot q \quad w_r = 100, w_e = 25$$

$$q = \sigma\left(-\frac{\log_{10} \varepsilon - c_0}{s}\right), \quad \varepsilon = \sqrt{\frac{\text{tr}((\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}})^+)}{n}}, \quad c_0 = 1.35, s = 0.70.$$

- $m = \sum_{ij} A_{ij}$ is the edge count; $\text{rank } \mathbf{B}$ comes from one thin SVD per step.
- $m c_{\max}$ is “the rank this many edges could at best have carried”, so both numerators are ranks and the score is dimensionless.
- A maximally informative edge gains $(w_r - w_e)c_{\max}/\text{rank}_K$ and pruning a redundant one gains $w_e c_{\max}/\text{rank}_K$: an exchange rate of exactly 3:1 in every domain, at every n . The optimum is $w_r - w_e = 75$.
- ε is **shape_err**, read off the rank_K nonzero eigenvalues of $\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}}$; lower is better, so q rises as ε falls. It is a *slope*, $d(\text{error})/d\sigma$, so its unit is rad^{-1} and it is not evaluated at $\sigma = 1$: see §2.1 for what it means and where it is valid.
- c_0, s are fixed constants, not per-episode quantities: no reference construction is built. **rigidity.SHAPE_ERR_EXPONENT** (default 0) optionally subtracts $\alpha \log_{10} n$ from the sigmoid argument, which makes φ ’s ceiling n -invariant. Re-derive all three with **tools/shape_error_scale.py**.
- $\kappa = \text{stiffness_kappa}$ scales the conditioning term. The term is bounded in $[0, \kappa \cdot \text{one edge})$, so κ is an *upper bound* on how many edges could ever be traded for conditioning, and it is a very loose one: see §2.2. $\kappa = 0$ reproduces the rank-only score exactly.
- The term is gated on rigidity, so becoming rigid is never punished.

Earlier score types, kept so archived runs replay

type	conditioning term	per-episode cost
WeightedNormalized	$\kappa (\text{one edge}) \sigma(\log_{10}(\lambda/\lambda_{\text{ref}})/0.75)$	3 greedy constructions
WeightedNormalizedSpectral	same, functional $\in \{\lambda, \text{tr}, \log \det\}$	3 greedy constructions
Weighted	$20 \text{rank } \mathbf{B} - 10m$, none	none

λ is the rigidity eigenvalue, the first nonzero eigenvalue of $\mathbf{B}^\top \mathbf{B}$. It has no fixed scale, which is why those two need λ_{ref} , the median λ of `stiffness_ref_samples` greedy constructions on the episode’s own poses. One construction rebuilds $\mathbf{B}$ once per candidate edge. `Weighted` does not transfer across domains: a rank-adding edge is worth +30 in $\mathbb{R}^3$ and +10 in $\mathbb{R}^2$.

2.1 What shape_err is

Each bearing is a unit vector, so its noise lives in the tangent plane. With $z_k = \text{normalize}(b_k + \sigma P(b_k)\varepsilon_k)$ and $\mathbf{B}$ the Jacobian of the bearing map (checked against central differences in all five domains), the chain is

$$\underbrace{\mathbf{B}^\top \mathbf{B} / \sigma^2}_{\text{Fisher}} \longrightarrow \underbrace{\sigma^2 (\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}})^+}_{\text{Cramér–Rao covariance}} \longrightarrow \mathbb{E}[\|\delta \hat{\chi} - \delta \chi\|^2] = \sigma^2 a_{\text{opt}}, \quad a_{\text{opt}} = \text{tr}((\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}})^+).$$

So a_{opt} is not a proxy for the error, it *is* the expected squared error per unit σ^2 ; “A-optimality” is only what the experiment-design literature calls $\text{tr} \Sigma$. Dividing by n and taking the root gives the RMS per-agent state error, position in formation radii and attitude in radians: $\sigma \varepsilon$ with $\varepsilon = \sqrt{a_{\text{opt}}/n}$.

The pseudo-inverse is what makes it a *shape* error. $\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}}$ is singular and its kernel is exactly the trivial variation set (Michieletto Thm. 1): translation, uniform scale, and the rotations the frames absorb, of dimension $6n - \text{rank}_{\mathcal{K}}$. The $^+$ inverts on the row space and returns zero there, so the sum runs over the identifiable directions only and the unobservable ones are never charged for. There is no separate gauge-quotient step. Measured at $n=6$ SE(3): $6n = 36$, $\text{rank}_{\mathcal{K}} = 29$, a 7-column gauge basis with $\|\tilde{\mathbf{B}} Z_{\mathcal{K}}\| = 7.0 \times 10^{-15}$ and the seven smallest eigenvalues at 10^{-15} .

σ is the per-axis standard deviation, not the angle. The tangent plane has two independent components each of standard deviation σ , so the RMS angle between the true and the noisy bearing is $\sigma\sqrt{2}$. Measured 1.40–1.42 over four decades. The Fisher information above is written in the same per-axis convention, so every validation is self-consistent; what needs the $\sqrt{2}$ is relating a noise level to a sensor specification. `--noise-sweep 1` applies a per-axis σ of one degree and an RMS angle of 1.41 degrees.

Where it holds. The whole construction is a linearisation. Measured over the ratio of the recovered error to the predicted one, it is exact while the predicted *relative* error $\sigma \varepsilon$ stays below roughly 0.2, and degrades above:

network	ε	0.06°	0.57°	1.72°	5.73°
$n=6 \mathbb{R}^3$	3.6	1.02	1.02	0.99	1.01
$n=6 \text{ SE}(3)$	6.2	1.00	1.01	1.07	0.96
$n=6 \mathbb{R}^2$	5.6	0.96	0.96	0.96	0.95
$n=8 \mathbb{R}^3$	29.4	1.06	0.80	0.61	0.18
$n=10$ mixed	22.5	0.95	1.04	0.87	0.41
$n=10$ mixed, ill conditioned	693.7	0.12	0.03	0.02	0.01

The rows break at a common $\sigma \varepsilon$, not at a common σ . For a typical ε of 5 to 30 that is σ below about 0.4° to 2°. Quote it at an operating point rather than at one radian: $\varepsilon = 8$ means one degree of per-axis bearing noise displaces the shape by about 14% of its own size.

2.2 What κ actually trades

Over rigid graphs the score collapses to

$$\varphi = 100 - \text{one edge} \cdot (m - \kappa q),$$

so the φ -optimal graph at a given κ is the one minimising $m - \kappa q$, and the optimum moves from m to $m + 1$ edges exactly when

$$\kappa (q^*(m + 1) - q^*(m)) > 1, \quad \text{i.e.} \quad \kappa > 1/\Delta q.$$

Because $q \in (0, 1)$, κ bounds the trade at κ edges. The realised trade is $\kappa \Delta q$, and Δq is small.

Measured by enumerating (m, q) for every graph and reading the κ curve off it exactly: exhaustive at $n = 4, 5$; at $n \geq 6$ the best graph per edge count is approximated by random restarts plus single-edge-swap hill climbing, which lower bounds q^* and so upper bounds the κ threshold. 46 networks over $n \in \{4, 5, 6, 8, 10\}$ and six compositions:

	range over 46 networks	typical
Δq per extra edge	0.005 – 0.041	0.015
$\Delta \log_{10} \varepsilon$ per extra edge	0.027 – 0.187 decades	0.07
κ needed to buy <i>one</i> edge	24 – 208	≈ 65
extra edges bought at $\kappa \leq 20$	0, in every network	
extra edges bought at $\kappa = 100$	0 – 3	

So at the $\kappa = 2$ the configs use, the optimum spends **no** extra edges at all. Part of the gap is that $q(m_{\text{req}})$ sits at 0.76–0.85 rather than at the sigmoid’s centre, since c_0 was calibrated on greedy constructions and an exhaustive optimum is better conditioned than one; a perfectly centred sigmoid would raise dq/dg from 0.23 to at most 0.357 and so cut the thresholds by about $1.6\times$, leaving them an order of magnitude above 2.

What κ does instead is break the tie that $\kappa = 0$ leaves. At $\kappa = 0$ every rigid graph on m_{req} edges scores identically, and those graphs are not equivalent at all. Exhaustively, over all rigid graphs at exactly m_{req} edges:

configuration	rigid graphs	worst/best ε	q range
$n=4 \mathbb{R}^3$	48	$3.1\times$	0.686–0.814
$n=4 \mathbb{R}^2$	192	$8.4\times$	0.596–0.847
$n=4 \text{SE}(3)$	108	$77.6\times$	0.201–0.790
$n=4 \mathbb{R}^3 \times S^1$	302	$610\times$	0.071–0.803
$n=5 \mathbb{R}^3$	4480	$9.3\times$	0.543–0.825
$n=5$ mixed	18648	$7079\times$	0.014–0.782

κ is therefore a *selector among equally sparse graphs* over the whole usable range, not a sparsity-versus-conditioning exchange rate. That is consistent with the greedy sweep, where edge count is flat to within 0.08 of an edge from $\kappa = 0$ to 4 while stiffness moves $20\times$.

3 Observation (obs_type = "Dict")

Five tensors. Widths below are per node or per ordered pair; bracketed flags are the env config keys that switch a block on. Order matches `build_dict_obs`’s concatenation.

node_features, $(n, \cdot)$

channel	w	definition
domain	5	one-hot over $(\mathbb{R}^2, \mathbb{R}^2 \times S^1, \mathbb{R}^3, \mathbb{R}^3 \times S^1, \text{SE}(3))$
degree	2	$(\deg_i^{\text{in}}, \deg_i^{\text{out}})/(m_{\text{req}}/n)$, i.e. relative to the mean degree of a minimal graph. Dividing by $n - 1$ instead introduces a $1/n$ trend
closeness	1	$\frac{r_i}{\sum_j d_{ij}} \cdot \frac{r_i}{n-1}$, r_i reachable nodes, d from Floyd–Warshall [graph_features]
eigenvector	1	power iteration on $A^\top$, unit norm [graph_features]
node_between	1	Brandes betweenness [graph_features]
rigidity_glob	3	$(\frac{\text{rank}_{\mathcal{K}} - \text{rank } \mathbf{B}}{\text{rank}_{\mathcal{K}}}, \frac{m}{m_{\text{req}}}, \not\llbracket \text{IBR} \rrbracket)$, tiled identically on every node [rigidity_global]
quality	1	the same q the reward uses, tiled; 0 while flexible [rigidity_quality]
node_freedom	1	$\ F_i\ $ divided by its RMS over nodes, where F is an orthonormal basis of $\ker \tilde{\mathbf{B}} \ominus \ker \tilde{\mathbf{B}}_{\mathcal{K}}$ (the non-trivial flex) and F_i is node i 's 6 rows. Zero once rigid [rigidity_flex]
node_slack	2	$(\ v_i^{\text{pos}}\ , \ v_i^{\text{att}}\)$, each divided by its own mean over nodes, with v the eigenvector of $\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}}$ at the rigidity eigenvalue. Zero while flexible [rigidity_stiffness]

edge_features, $(n, n, \cdot)$ over *all* ordered pairs

channel	w	definition
bearings	3	$b_{ij} = R_i^\top \hat{p}_{ij}$ for every ordered pair, edge or not. <code>include_candidate_bearings=False</code> reverts to existing edges only, same shape
edge_exists	1	$A_{ij} \in \{0, 1\}$
edge_between	1	Brandes edge betweenness [graph_features]
reciprocity	1	$A_{ij} \wedge A_{ji}$
common_nbrs	1	$(A^2)_{ij}/(m_{\text{req}}/n)$
add_independence	1	$\ b_{ij}Z\ _F/\ b_{ij}\ _F \in [0, 1]$, Z an orthonormal basis of $\ker \tilde{\mathbf{B}}$. Zero exactly on the pairs that would add no rank; zero everywhere once rigid [rigidity_flex]
pair_max_rank	1	c_k/c_{max} from the <i>complete</i> graph, so a candidate reads its own value rather than 0. Constant in any homogeneous network [rigidity_edge]
add_rank	1	$\text{rank}(b_{ij}Z)/c_{\text{max}}$, the exact rank adding $i \rightarrow j$ would gain. Threshold <code>add_independence</code> $> 10^{-6}$, which sits mid-way in an eight-decade gap [rigidity_edge]
add_stiffness	1	$\ b_{ij}v\ $ divided by its own mean over pairs; first-order $\Delta\lambda \approx \ b_{ij}v\ ^2$. A ranking prior, not an oracle. Zero while flexible [rigidity_stiffness]
remove_rank	1	rank lost by deleting $i \rightarrow j$, over c_{max} : the number of unit eigenvalues of the leverage block $H = b(\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}})^+ b^\top$. Exact. Zero on non-edges [rigidity_removal]

channel	w	definition
<code>remove_stiffness</code>	1	$1 - \lambda(\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}} - b^\top b) / \lambda(\tilde{\mathbf{B}}^\top \tilde{\mathbf{B}}) \in [0, 1]$, exact via the rank-3 downdate; 1 when removal breaks rigidity. Zero on non-edges <code>[rigidity_removal]</code>

b_{ij} is the $3 \times 6n$ block edge $i \rightarrow j$ would contribute to $\tilde{\mathbf{B}}$; it is never materialised, `candidate_gain` expands $b_{ij}Z = D_p(S_j Z_j - S_i Z_i) - D_a P_i Z_i$ into batched products over all pairs and reads norm and rank off the 3×3 Gram matrix.

Remaining tensors

key	shape	definition
<code>coord_features</code>	$(n, 3)$	positions centred on the centroid, divided by L
<code>adj</code>	(n, n)	adjacency; this is what the model’s action mask reads
<code>selection</code>	(n)	pointer state, <code>SelectNodesSequentially</code> only

4 Invariance

The task is defined up to a similarity transform, so the observation and φ should be too. Measured by transforming the network and diffing each channel, both regimes (flexible and rigid), $\mathbb{R}^3$ / $\text{SE}(3)$ / mixed, n up to 100.

	translation	rotation	uniform scaling
φ	exact	exact	exact
every rigidity channel	$\leq 10^{-11}$	$\leq 10^{-11}$	$\leq 10^{-11}$
<code>degree, adj, reciprocity, common_nbrs</code>	exact	exact	exact
<code>bearings</code>	exact	moves in $\mathbb{R}^2, \mathbb{R}^3$	exact
<code>coord_features</code>	exact	equivariant by design	exact

- `bearings` rotate in the frameless domains because $R_i = I$ there, so `get_bearing` returns a global-frame vector. Exact to 10^{-15} in $\mathbb{R}^2 \times S^1$, $\mathbb{R}^3 \times S^1$, $\text{SE}(3)$. `rotation_augmentation` is the mitigation.
- `coord_features` rotate deliberately. GINE never receives them; EGNN reads them only through $\|x_i - x_j\|^2$, which is invariant.
- Scale invariance of the three stiffness channels requires that v , Z and the removal downdate all come from $\tilde{\mathbf{B}}$ rather than from $\mathbf{B}$. Mapping a null *space* into scaled units afterwards is valid; applying the same map to the single eigenvector v is not.
- Channels normalised by their own mean or RMS (`node_freedom`, `node_slack`, `add_stiffness`) encode *which* node or pair is soft, never how soft in absolute terms.

5 Scale under a per-episode domain draw

With `random_domains`, the composition is redrawn every `reset`, so rank_K , $c_{\max}$ and m_{req} move between episodes and with them the size of one reward step. Measured over 4000 uniform draws per n :

n	one edge, median [p10–p90]	p90/p10	φ optimum p90/p10
8	1.923 [1.613–2.381]	1.48	1.015
16	0.926 [0.806–1.042]	1.29	1.007
32	0.446 [0.410–0.490]	1.20	1.003
64	0.220 [0.207–0.234]	1.13	1.002
100	0.140 [0.133–0.147]	1.11	1.001

- The spread *narrows* with n : $\text{rank}_{\mathcal{K}}$ is a sum of n i.i.d. per-agent DOFs, so its relative spread falls as $1/\sqrt{n}$. The corner-to-corner range is a flat $2.0\times$ at every n ; uniform draws occupy a shrinking part of it.
- φ 's optimum is effectively constant, which is what a value-based method needs: with a potential-based reward the return telescopes to $\varphi(s_T) - \varphi(s_0)$.
- $c_{\max} = 1$ requires every agent planar: 0.05% of draws at $n = 8$, none above.
- Within one episode the realised rewards already span -3 to $+3$ times one edge. Measured across 120 episodes, the within-episode standard deviation exceeds the between-episode one by $2\text{--}3.8\times$, and switching from a fixed mix to a random one raises the between-episode figure from 0.434 to 0.507 at $n = 10$ and not at all at $n = 20$.
- Across *sizes* the step size does move by an order of magnitude (1.92 at $n = 8$ to 0.109 at $n = 128$), since one edge is $w_e c_{\max} / \text{rank}_{\mathcal{K}}$ and $\text{rank}_{\mathcal{K}}$ grows linearly in n .